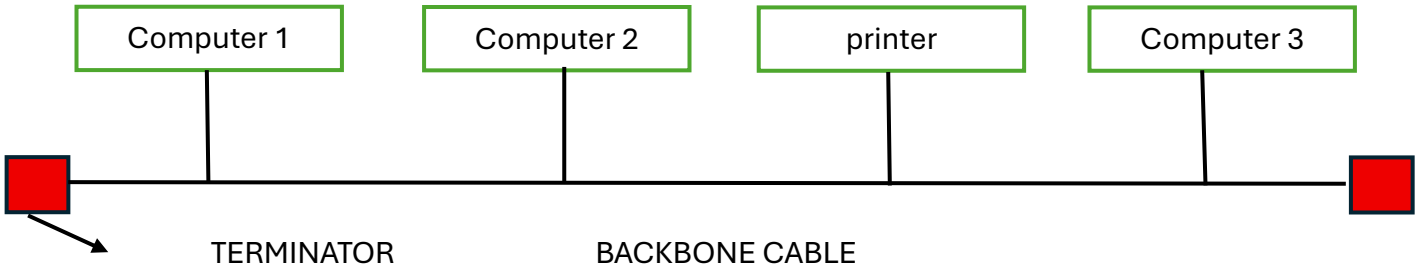
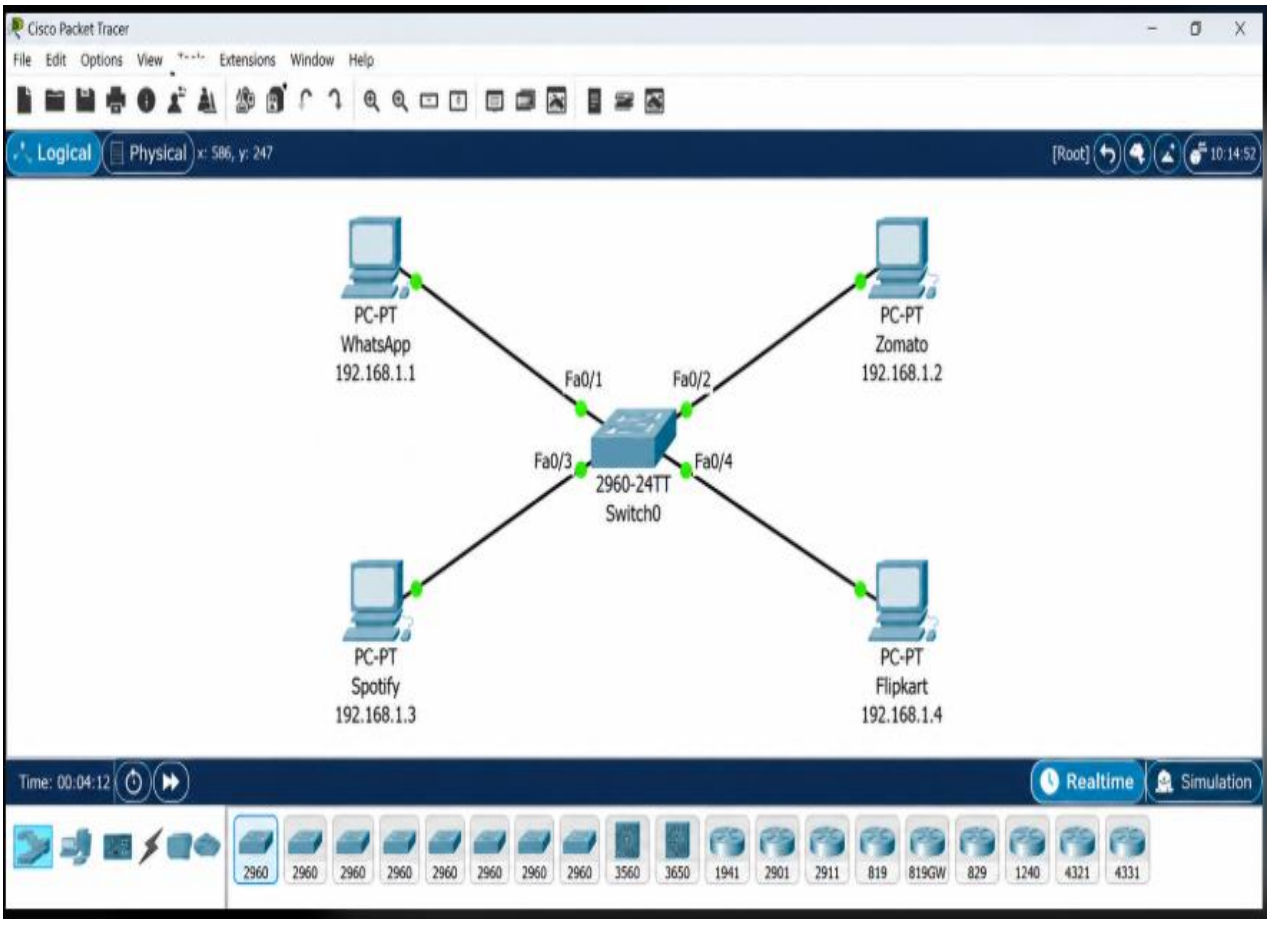
1. Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.



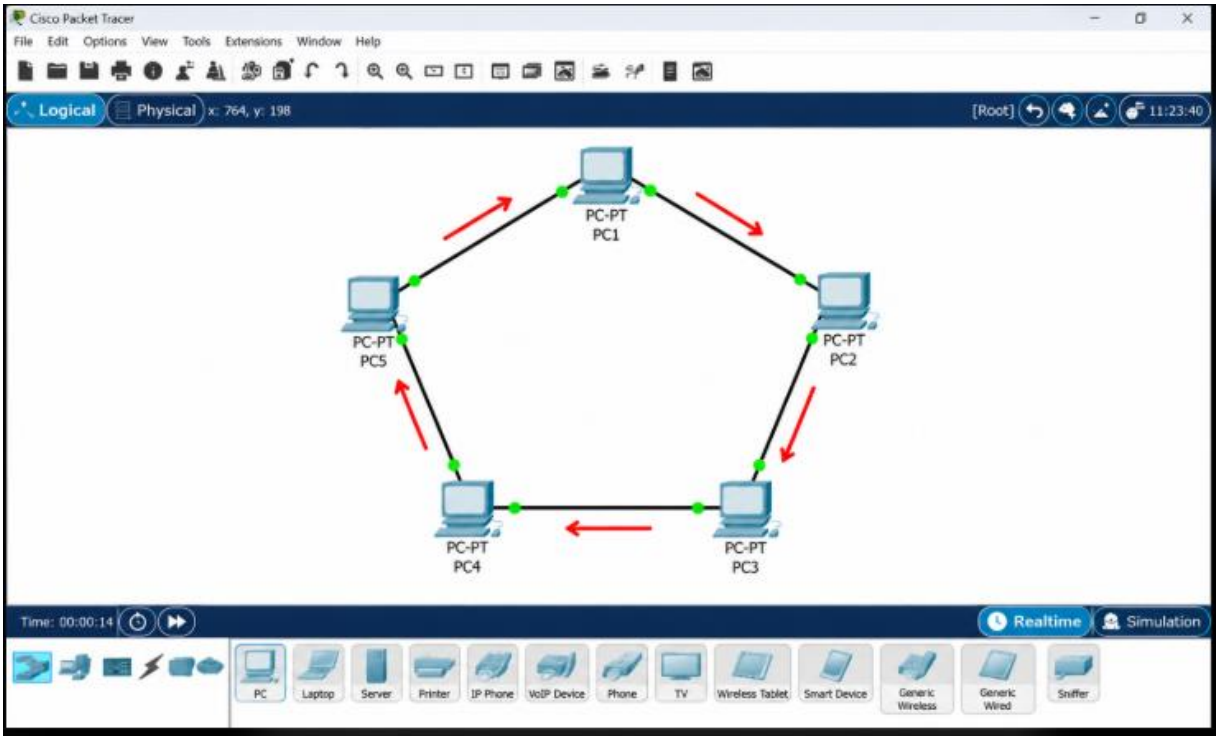
Real world example: A bus topology can be use in an older office network where a few computrers are connected using a single co-axial cable.

2. In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.



3. Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.

Hint: Use arrows to show the direction of data flow.



Data travels from the one device to the next device in a fixed direction usually clockwise or counter clockwise. Each device receives the data, checks if it is meant for it, and if not, forwards it to the next device in the ring.

4. Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

Topology	Advantages (2)	Disadvantages (2)
Star Topology	1. Easy to install, manage, and troubleshoot. 2. Failure of one PC or its cable does not affect other devices.	1. If the central switch fails, the entire network stops working. 2. Requires more cables than bus topology.
Mesh Topology	1. Highly reliable because multiple paths are available for data transmission. 2. Failure of one connection usually does not affect the entire network.	1. Expensive to install because it requires more cables and connections. 2. Complex to configure and maintain, especially in large networks.